



ShortLink Pro

Functional Documentation

A concise functional overview of the user features, administration capabilities, technical scope, requirements, installation process, and support information for ShortLink Pro.

Document Scope

This document describes what each product area is intended to do. It is a functional overview for product owners, administrators, support teams, and implementation stakeholders. It is not a low-level API specification.

Product name: ShortLink Pro

Document type: Functional and implementation overview

Primary audience: buyers, administrators, and implementation teams

Product Overview

ShortLink Pro is a self-hosted URL shortener and link management platform for creating branded short links, organizing campaigns, connecting custom domains, tracking link performance, and operating a subscription-based SaaS workflow from a modern dashboard.

The application is suitable for marketing teams, agencies, SaaS products, internal business tools, and website owners who need more control than public link shortening services can provide.

Core Capabilities

Link Management Create short URLs one at a time or in bulk, manage aliases, passwords, expirations, and redirects. - Generated or custom aliases - Password-protected links - Expiration dates for campaigns	Organization Group related links into Workspaces for campaigns, clients, departments, or topics. - Color-coded Workspaces - User-owned organization structure - Fast filtering and management
Branding Use custom domains to build recognizable and trustworthy short links. - Custom branded domains - Domain-based link ownership - Improved link confidence	Analytics Understand traffic and audience behavior with detailed statistics. - Clicks, devices, browsers, platforms - Locations and traffic sources - Campaign performance insight

Requirements

ShortLink Pro can be installed with the included Docker environment or on a compatible PHP hosting/server stack. Docker is recommended for predictable local deployment.

PHP	PHP 8.4 or compatible runtime with required extensions.
PHP Extensions	BCMath, cURL, DOM, Fileinfo, GD, Intl, JSON, Mbstring, OpenSSL, PDO, PDO MySQL, Tokenizer, XML, and Zip.
Database	MySQL 8.x or a compatible MySQL/MariaDB server.
Web Server	Apache2 with the public directory configured as the document root, or another supported web server.
Package Managers	Composer 2 and Node.js/npm when installing or rebuilding dependencies/assets manually.
Docker Setup	Docker and Docker Compose v2 when using the included container environment.
Permissions	The application must be able to write to storage, bootstrap/cache, and the .env file during installation.

Recommended Docker Ports

- Application URL: `http://localhost:8090`
- Host MySQL port: `127.0.0.1:3308`
- Docker network MySQL endpoint: `mysql:3306`

Installation Guide

The following instructions describe the recommended Docker-based installation flow. They assume that the project files have been copied to the target machine and that Docker is already installed.

1. Prepare the environment file

Copy the Docker environment template if the .env file does not exist yet.

```
cp .env.docker .env
```

2. Build and start the containers

```
docker compose up -d --build
```

3. Install dependencies and prepare the application

The Docker entrypoint can install Composer dependencies, generate the application key, create the storage symlink, install npm dependencies, and build assets automatically. If manual execution is required, run:

```
docker compose exec app composer install
docker compose exec app php artisan key:generate
docker compose exec app php artisan storage:link
docker compose exec app npm install
docker compose exec app npm run build
```

4. Run database migrations

```
docker compose exec app php artisan migrate
```

5. Open the application

```
http://localhost:8090
```

6. Useful maintenance commands

```
docker compose ps
docker compose logs -f app
docker compose exec app php artisan optimize:clear
docker compose exec app php artisan test
```

User Features

User-facing features focus on creating, organizing, protecting, promoting, and analyzing links. They help users build measurable campaigns without leaving the platform.

LINKS Shorten links one at a time or in bulk and manage the full link lifecycle. - Create generated or custom aliases - Protect selected links with passwords - Run campaigns with expiration dates	WORKSPACES Organize links into Workspaces so users can manage campaigns, clients, or topics more easily. - Group related links - Use colors to recognize Workspaces at a glance
DOMAINS Brand short links with custom domains and make URLs more recognizable and trustworthy. - Attach user-owned domains - Increase trust and click-through potential	STATS Analyze link performance and understand the audience behind each click. - Track location, language, browser, platform, device, source, and social media - Use insights to improve campaigns
GEOGRAPHIC TARGETING Redirect visitors to different destination URLs based on their geographical location. - Create country-specific routing rules - Send regional audiences to localized pages or offers	PLATFORM TARGETING Redirect visitors based on the platform or device they are using. - Create rules for mobile, desktop, or platform categories - Send users to the most relevant destination
SHARING Share created links through common channels without leaving the application workflow. - Share by email, QR code, X, Facebook, Reddit, Pinterest, and LinkedIn - Use quick actions after creating or editing links	

Admin Features

Administration features provide operational control over website settings, subscriptions, users, content, and objects created by users.

DASHBOARD View current website activity from a single control panel. - Monitor high-level counters - Review recent links, top views, and activity summaries	SETTINGS Configure global website behavior and operational settings. - Registration, email, payment, legal, invoice, captcha, contact, and shortener options - Update settings without changing application code
PLANS Manage predefined or custom subscription plans and adapt them to business needs. - Configure plan visibility and pricing references - Control feature limits for links, Workspaces, domains, stats, targeting, and API access	SUBSCRIPTIONS View user subscriptions and assign manual subscriptions when needed. - Inspect status, plan, billing period, and owner - Create manual subscriptions for administrative access
USERS Manage website accounts and review user-owned objects. - View, edit, delete, restore, or suspend users - Open related links, Workspaces, domains, and subscriptions	WORKSPACES View, edit, or delete Workspaces created by users. - Inspect owner and metadata - Open all links that belong to a Workspace
DOMAINS View, edit, or delete custom domains added by users. - Inspect owner and fallback pages - Open all links connected to a domain	PAGES Create and edit informational, footer, legal, and website content pages. - Manage content from the admin panel - Keep static pages editable without code changes
LANGUAGE SWITCHING Switch the interface language in the admin panel and user dashboard. - English is the default language - Supports English, German, Russian, French, Spanish, Chinese, and Arabic	

Technical Overview

ShortLink Pro is structured as a Laravel application with a modern AdminLTE 4 administration interface, Docker support, and a layered code organization for maintainability.

Backend	Laravel 13, PHP 8.4, service layer, repositories, DTOs, form requests, enums, observers, and models.
Database	MySQL with migrations and seeders for application data.
Frontend	AdminLTE 4, Bootstrap-based interface, Vite asset build flow, responsive admin and user dashboards.
Environment	Docker Compose setup with Apache2, PHP 8.4, and MySQL.
Security	Authentication, registration, password reset workflows, validation requests, and reCAPTCHA v3 support.
Internationalization	Language files and interface switching for supported languages.

Included Package Materials

- Laravel application source code
- Admin panel and user dashboard source code
- Database migrations and seeders
- Docker configuration
- Public assets and compiled frontend assets
- README deployment documentation
- Functional PDF documentation
- Marketplace icon and preview images

Author and Customization

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Customization Notice

On customer request, I can adapt the script to specific project requirements. Various additional features can be added or existing functionality can be improved or extended according to the customer's needs.

This makes ShortLink Pro suitable both as a ready-to-use product and as a flexible base for custom SaaS, marketing, analytics, or internal link management workflows.

Thank you for choosing ShortLink Pro.